



National Kidney Month Clinical Update

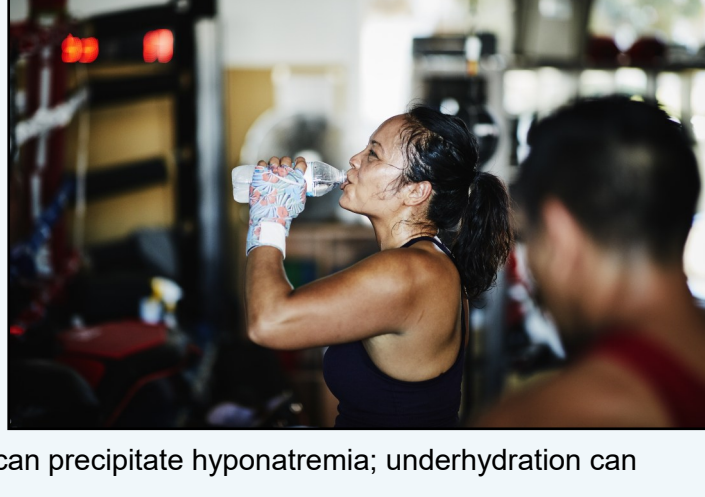
Kidney Health Management and Monitoring for Virtual Care Providers

March is National Kidney Month, an opportunity to reinforce best practices in kidney health promotion, early detection, and safe monitoring. Virtual care visits are essential opportunities to apply these principles, where history-taking and appropriate lab surveillance are critical.

General Hydration Recommendations

Hydration guidance should be individualized.

- **Healthy Adults:** Encourage fluid intake sufficient to maintain light yellow urine. For most adults, this approximates **2–3 liters/day total fluid intake**, but varies by body size, activity, and climate.
- **Chronic Kidney Disease (CKD) Patients:** Avoid blanket “push fluids” advice. In CKD stages 3–5 or in heart failure, fluid intake should be guided by volume status and cardiology/nephrology recommendations.
- **Nephrolithiasis History:** Aim for urine output $\geq 2\text{--}2.5$ L/day to reduce recurrence risk.
- **Caution in Elderly Patients:** Overhydration can precipitate hyponatremia; underhydration can worsen AKI risk.

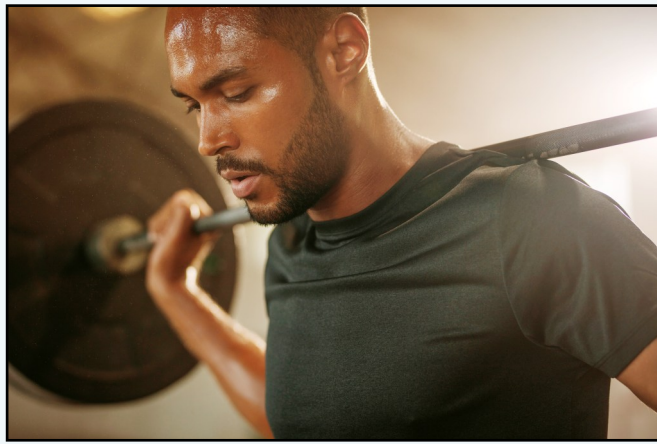


Virtual Care Pearl: Ask about urine color, orthostatic symptoms, weight changes, edema, and home BP trends to approximate volume status. Listen and counsel empathetically if you elicit history related to hydration myths such as “a gallon a day.”

Creatine Supplementation: Current Evidence

Creatine remains popular among athletes and adults for muscle mass preservation.

Key Points for Providers



- Creatine causes small increases in **serum creatinine** due to conversion to creatinine—not necessarily due to kidney injury.
- In healthy individuals with normal renal function, there is **no consistent evidence of kidney damage** when used at recommended doses (typically 3–5 g/day after loading phase in studies up to five years).
- Other markers of renal function—including urea, electrolytes, proteinuria, and albuminuria—remain unchanged with supplementation.

- Risk increases in:
 - Underlying CKD
 - Concurrent nephrotoxic medications
 - Dehydration
 - High-dose or unregulated supplement use

Clinical Approach in Virtual Care

- Clarify dose.
- Establish **baseline creatinine/eGFR** before initiation
- While safe in healthy populations, **creatine supplementation is NOT recommended for individuals with pre-existing renal disease or risk factors for renal dysfunction** (e.g., diabetes, hypertension, reduced GFR).

Advise discontinuation and in-person evaluation if:

- Rapid creatinine rise
- Reduced urine output
- Edema
- Hypertension worsening

Monitoring Labs for Patients on Diuretics and ACEi / ARB

ACE inhibitors and ARBs are foundational for hypertension, heart failure (HF), and CKD with albuminuria, but they require monitoring.

ACE Inhibitors / ARBs

Selection: Do not initiate if baseline potassium is greater than 4.5.

Baseline Testing: Creatinine, potassium, blood pressure

After Initiation or Dose Change: Repeat creatinine and potassium in **1–2 weeks**. Repeat blood pressure in 1-2 weeks.

Expected Changes

- Up to **30% increase in creatinine** from baseline may be acceptable.
- Mild potassium increase (≤ 5.5) may be monitored closely.

Red Flags Requiring Urgent Evaluation

- Creatinine rise $>30\%$
- Potassium ≥ 5.6
- Symptomatic hypotension

Diuretics

Loop Diuretics (e.g., Furosemide)

- Monitor creatinine, sodium, and potassium within 1–2 weeks after initiation/dose change.
- Watch for hypokalemia, hyponatremia, and rising creatinine.

Thiazides

- Monitor creatinine, sodium, and potassium
- Use with caution in those with gout; monitor uric acid
- Watch for hyponatremia in older adults.

Spirolactone

- Monitor sodium, potassium, and creatinine
- Do not initiate if baseline potassium > 4.5

Virtual Assessment Pearls

- Daily weights (HF patients)
- Dizziness/orthostasis
- Muscle cramps
- Confusion (possible hyponatremia)



Virtual Workflow Tip: After medication initiation or changes, order labs as future orders.

Monitoring Chronic Kidney Disease

CKD is defined as **eGFR <60 mL/min/1.73m² for ≥ 3 months**, or persistent albuminuria. Both GFR and urine albumin-to-creatinine ratio (ACR) are needed to provide staging and calculate risk of progression.

Other Core Labs

- Electrolytes
- Urine ACR and protein-to-creatinine annually
- Hemoglobin and hematocrit (anemia)
- Phosphorus, calcium, vitamin D, PTH in advanced CKD

Medication Review

- Avoid NSAIDs when possible.
- Adjust dose for renally-cleared drugs. Recognize that some medication guidance is given as creatinine clearance (use **.CRCLCALC**), which is different than GFR (use **.LASTGFR**).
- Reinforce ACEi / ARB use in albuminuric CKD, unless contraindicated.

Virtual Workflow: Use **.CKDRISK5YEAR** to calculate a patient’s 5-year risk of progression to renal failure requiring dialysis. If risk is greater than 5%, a nephrology referral is recommended.

Differentiating CKD v. Acute Kidney Injury (AKI)

Historical Clues for CKD

- Known long-standing hypertension or diabetes
- Prior abnormal labs
- Gradual fatigue
- Anemia history
- Long-term NSAID exposure
- No acute symptoms

Historical Clues for AKI (Urgent In-Person Evaluation Needed)

Rapid Onset (Within 1-2 Weeks) With

- Recent dehydration (GI illness, heat exposure)
- Sepsis symptoms (fever, chills)
- New medications (NSAIDs, contrast, antibiotics)
- Reduced urine output (<0.5 mL/kg/hr or patient reports “barely urinating”)
- Sudden edema
- Flank pain
- Gross hematuria
- Severe uncontrolled hypertension
- Confusion (possible uremia)



Red Flags Requiring Same-Day Evaluation

- Anuria or near-anuria
- Chest pain, dyspnea (volume overload)
- Potassium ≥ 5.6 (if labs known)
- Severe weakness (possible hyperkalemia)
- New neurologic changes (uremia)

When uncertain in virtual care, or if concerning features are present, direct patient to urgent care/ED.

References

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